

Weekly Assignment: Formal Languages & Regular Expressions (Epp 12.1)

Complete each of the following problems. Show all work and justify your answers. Recall the order of precedence for regular expression operations: $*$ (highest), then concatenation, then $|$ (lowest). Also recall the shorthand notations: character classes such as $[0-9]$, the $+$ operator ($r^+ = rr^*$), the $?$ operator ($r? = (\lambda|r)$), and $r\{n\}$ for exactly n repetitions.

Problem 1 (Epp 12.1 #2). Let $\Sigma = \{x, y\}$ be an alphabet.

- a. Let L_3 be the language consisting of all strings over Σ of length ≤ 3 in which all the x 's appear to the left of all the y 's. List the elements of L_3 between braces.
- b. List between braces the elements of Σ^4 , the set of all strings of length 4 over Σ .
- c. Let $A = \Sigma^0 \cup \Sigma^1$ and $B = \Sigma^3 \cup \Sigma^4$. Describe A , B , and $A \cup B$ in words.

Problem 2 (Epp 12.1 #11). Use the rules about order of precedence to eliminate the parentheses in the following regular expression:

$$(1(1^*)) \mid \left((1(0^*)) \mid ((1^*)1) \right)$$

Problem 3 (Epp 12.1 #23). Indicate whether the given strings belong to the language defined by the regular expression below. Briefly justify your answers.

Expression: $(x^*y|zy^*)^*$ Strings: $zyyxz$, $zyyzy$

Problem 4 (Epp 12.1 #38). Write a regular expression to define the set of all telephone numbers with the following format: three digits, then a hyphen, then three more digits, then a hyphen, and then four digits, where the first three digits are either 800 or 888 and the last four digits start and end with a 2.